

Chibuike Robinson Umeike

ML Research Scientist — Multimodal Foundation Models & Scientific AI

✉ chibuikeumeike@gmail.com 📞 +1-205-887-3455 🌐 crumeike.github.io
🐙 github.com/crumeike 🔗 linkedin.com/in/chibuike-robinson-umeike-b57b1a183

PROFESSIONAL SUMMARY

ML Research Scientist with expertise in large-scale deep learning, multimodal foundation models, and end-to-end ML workflow design for scientific applications. At Argonne National Laboratory, built end-to-end data pipelines processing **42,673 scientific articles** to generate structured image-text datasets, fine-tuned vision-language models achieving **135% performance improvement** and **96.6% hallucination reduction**, and developed **PtychoBench**, a benchmark for evaluating foundation model adaptation to X-ray ptychography data from the Advanced Photon Source. Experienced in multi-GPU training, scalable data pipelines, model optimization, and reproducible research on leadership-class computing systems. Published at NeurIPS and IEEE ISBI.

TECHNICAL SKILLS

- **ML/AI Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers, vLLM, LoRA, Scikit-Learn, OpenCV, VERL
- **Foundation Models:** LLaVA, Qwen, Llama, Vicuna – fine-tuning, RLHF/GRPO Alignment, LLM-as-judge evaluation
- **Multimodal & Vision:** Vision-language model fusion, CLIP, YOLO, RT-DETR, ResNet, SSD, image-text pair curation, figure-caption linking
- **Scientific Data:** X-ray ptychography (synchrotron/APS), biomedical imaging, heterogeneous document parsing, structured extraction from scientific literature
- **Development Tools:** Git, Weights & Biases, Plotly, Tesseract, NumPy, Pandas, Easyocr, LaTeX, PDFFigures, Marker
- **Production ML:** Multi-GPU training, ALCF HPC infrastructure, real-time inference pipelines, scalable data curation (171K+ samples)
- **Languages:** Python, C#, C++, Java, MATLAB, Bash

PROFESSIONAL EXPERIENCE

- **Argonne National Laboratory (U.S. Department of Energy)** Lemont, IL
• **Research Technical Aide** — *AI/ML for Scientific Computing* May 2024 – Aug 2025
 - **Foundation Models for Scientific Imaging (Summer 2025):** • Engineered domain-aware vision-language models (Qwen-2.5VL, Llama-3.2-Vision) using reinforcement learning (GRPO/PPO) for structured interpretation of X-ray ptychography data from the Advanced Photon Source. • Designed and released PtychoBench, a benchmark for evaluating foundation model adaptation in low-data physical science regimes. • Managed end-to-end pipeline from raw synchrotron data ingestion to structured model-ready representations, integrating domain expert feedback. • *Outcome: First-author paper at NeurIPS 2025 AI4MAT Workshop; NeurIPS Travel Grant recipient.*
 - **Multimodal Knowledge Extraction from Scientific Literature (Summer 2024):** • Built end-to-end data curation pipeline processing 42,673 scientific articles, extracting and structuring 171K+ image-caption pairs and 165K+ figures for model training. • Fine-tuned LLaVA vision-language models (13B parameters) using LoRA on 50,882 image-text pairs, with figure-caption linking. • Developed LLM-as-judge evaluation framework for assessing factual accuracy of multimodal model outputs against ground-truth scientific data. • Achieved 135% performance improvement and 96.6% hallucination reduction over baselines. • *Outcome: Oral presentation at IEEE ISBI 2025 (≈20% acceptance rate); invited to Trillion Parameter Consortium (90+ institutions across 4 continents).*
 - **Infrastructure & Collaboration:** • Utilized ALCF leadership-class HPC infrastructure for large-scale distributed multi-GPU model training and evaluation. • Collaborated with Argonne Computing, Environment and Life Sciences Directorate. • Presented findings to DOE leadership, Argonne senior scientists, and the Trillion Parameter Consortium
- **The University of Alabama** Tuscaloosa, AL
• **Doctoral Researcher** — *AI for Infrastructure & Smart Cities* June 2021 – Present
 - **Real-Time Scientific Vision Pipeline:** • Designed end-to-end deep learning pipeline (YOLO, RT-DETR, ResNet) for automated structural damage assessment with ordinal supervision, processing **333 images/second**, demonstrating scalable real-time inference on heterogeneous image data. • *Outcome: 2nd place NSBE National Convention (15,000+ attendees), ASCE Editor's Choice Award.*
 - **Multi-Institutional Research Leadership:** • Led technical coordination for NIST Disaster and Failure Studies Program consortium (NIST, Johns Hopkins, Colorado State, South Alabama). • Delivered bi-weekly progress updates to federal stakeholders. • Project showcased at NIST AI Day 2024.

- **IoT & Heterogeneous Data Integration:** • First application of OIIE international standards to real-time multi-modal urban data (sensor networks, geospatial, temporal streams). • Processed 25,000+ heterogeneous data points into structured decision-support representations. • *Outcome: 2 first-place innovation awards in 2020 & 2022, TISDIC 2023 publication.*

• **VFD Tech Limited (VFD Group Plc)**

Lagos, Nigeria

• **IT Project Manager**

June 2020 – June 2021

- **Product Management/Platform Development:** • Managed 25+ technology projects; led second major release of V mobile banking app (1M+ Android downloads). • Oversaw Banking-as-a-Service platform serving 20+ corporate clients.

EDUCATION

- **University of Alabama — Dissertation Defended & Passed** Tuscaloosa, AL
PhD, Civil Engineering (AI/Computer Vision focus) — GPA: 3.85/4.0 Degree Conferred May 2026
 – Dissertation: *Automated Framework for Rapid Post-Disaster Infrastructure Assessment Using Deep Learning*
- **University of Alabama** Tuscaloosa, AL
MS, Computer Science — GPA: 3.85/4.0 Dec 2024
- **University of Alabama** Tuscaloosa, AL
MS, Civil Engineering — GPA: 4.0/4.0 Dec 2023

KEY PROJECTS & OPEN SOURCE

- **PtychoBench** (github.com/crumeike/PtychoBench): Benchmark for evaluating foundation model adaptation to X-ray ptychography. Domain-aware VLMs and LLMs (Llama-3.1, Llama-3.2-vision). Published **NeurIPS 2025 AI4MAT**.
- **LUCID** (github.com/crumeike/lucid-biomedical-vlm): End-to-end pipeline: 42K article ingestion, 171K image-caption curation, LoRA fine-tuning, LLM-as-judge evaluation. Published **IEEE ISBI 2025**.
- **TornadoNet** (github.com/crumeike/TornadoNet): Real-time building damage detection with ordinal supervision using YOLO/RT-DETR. Multi-institutional **NIST consortium project**.

SELECTED PUBLICATIONS (11 TOTAL, 8 FIRST-AUTHOR, 2 SOLE-AUTHOR)

- **R. Umeike**, T. Dao, S. Crawford, J. van de Lindt et al., "Architectural Insights for Post-Tornado Damage Recognition," *arXiv preprint arXiv:2602.14523*, 2026. [PDF]
- **R. Umeike**, N. Getty, X. Yin, Y. Jiang, "Adapting General-Purpose Foundation Models for X-ray Ptychography in Low-Data Regimes," *NeurIPS 2025 AI4MAT Workshop*, 2025.
- **R. Umeike**, N. Getty, F. Xia, R. Stevens, "Scaling Large Vision-Language Models for Enhanced Multimodal Comprehension in Biomedical Image Analysis," *IEEE ISBI 2025 (Oral)*, 2025.
- **R. Umeike**, T. Dao, S. Crawford, "State-of-the-Art Review of Resilient Smart Cities: Progress and Challenges," *Urban Governance*, vol. 5, no. 2, pp. 222-239, 2025.
- W. Wang, J. van de Lindt,...,**R. Umeike** et al., "Application of Multidisciplinary Community Resilience Modeling to Reduce Disaster Risk," *J. Performance of Constructed Facilities* (ASCE Editor's Choice), vol. 38, no. 3, 2024.
- **R. Umeike**, "Comparative Crime Analysis and Prediction Using Machine Learning Algorithms: Assessing the Tools and Addressing the Threats," *IEEE ICTAI 2023*, pp. 135-142, 2023.
- **Full publication list:** scholar.google.com/citations?user=IeJsfl8AAAAJ

HONORS & RECOGNITION

- **NeurIPS AI4MAT Travel Grant** - Competitive award for conference presentation (2025)
- **2nd Place, NSBE National Technical Research Exhibition** - 15,000+ attendees (2025)
- **ASCE Editor's Choice Award** - Awarded to < 2% of all 200,000+ ASCE publications (2024)
- **Argonne Sustainable Research Pathways Fellowship** - Competitive fellowship for placement in DOE (2024)
- **Invited Speaker, Trillion Parameter Consortium** - Global AI consortium, 90+ institutions, 4 continents (2024)
- **1st Place, UA Faculty & Staff Innovation Competition** – Led technical implementation of e-Flood IIoT system; \$5,000 First Prize, Division of Finance and Operations (2022)
- **Peer Review Service:** over 46 reviews for IEEE ISBI, IJDRR (Ranked #1 in emergency management), Engineering Applications of AI, ASCE, Wiley, Frontier & Elsevier journals
- **Media Features:** Argonne National Laboratory, ALCF, TPC Seminar Series, ASCE Civil Engineering Source